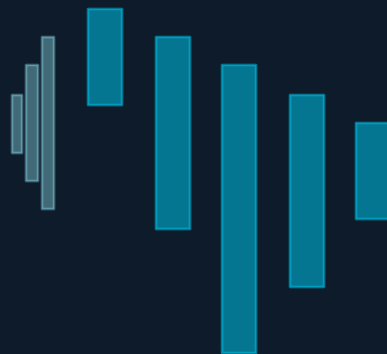




HearEdit

Speaker Diarization System

Milestone 2



Agenda

Milestone 2

01

Project Overview

02

Overall System Architecture

03

Model Setup and Preparation

04

API Design and Model Serving

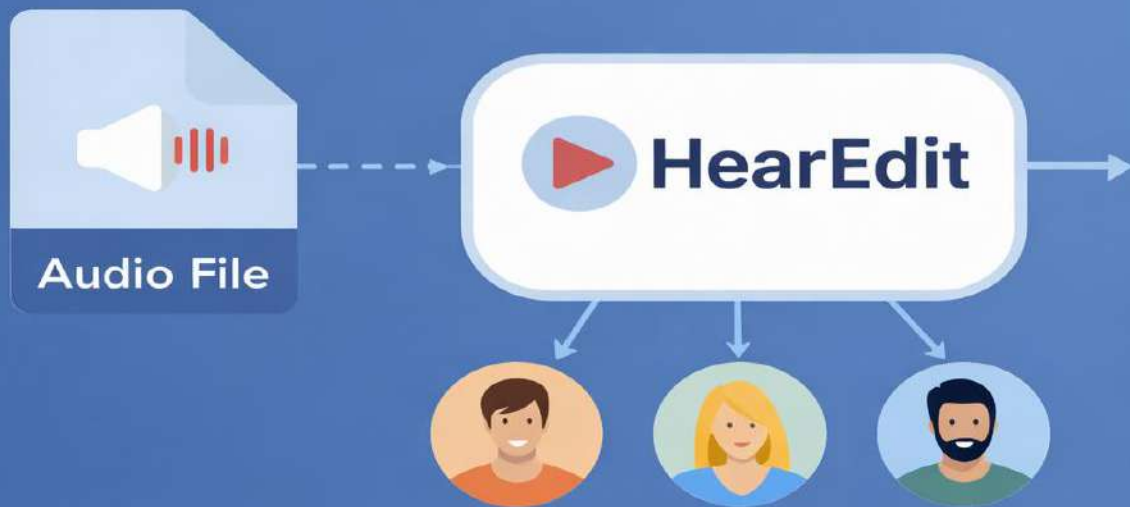
05

Deployment in the Cloud

06

Conclusion

1. Project Overview



Transcription



Speaker 1

Hey, are we still on for our meeting at 10 AM?



Speaker 2

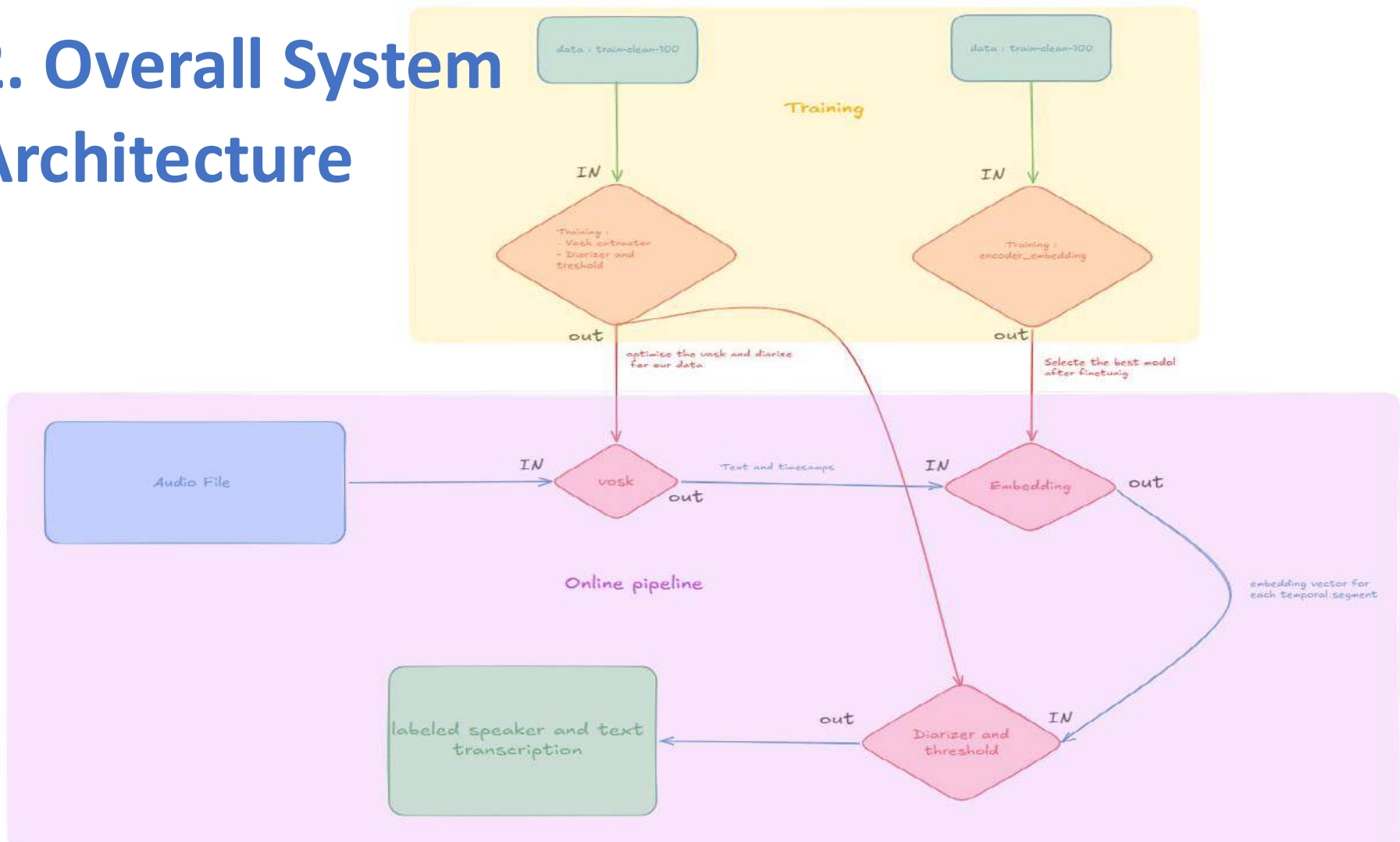
Yes, I'll be there. Looking forward to it!



Speaker 3

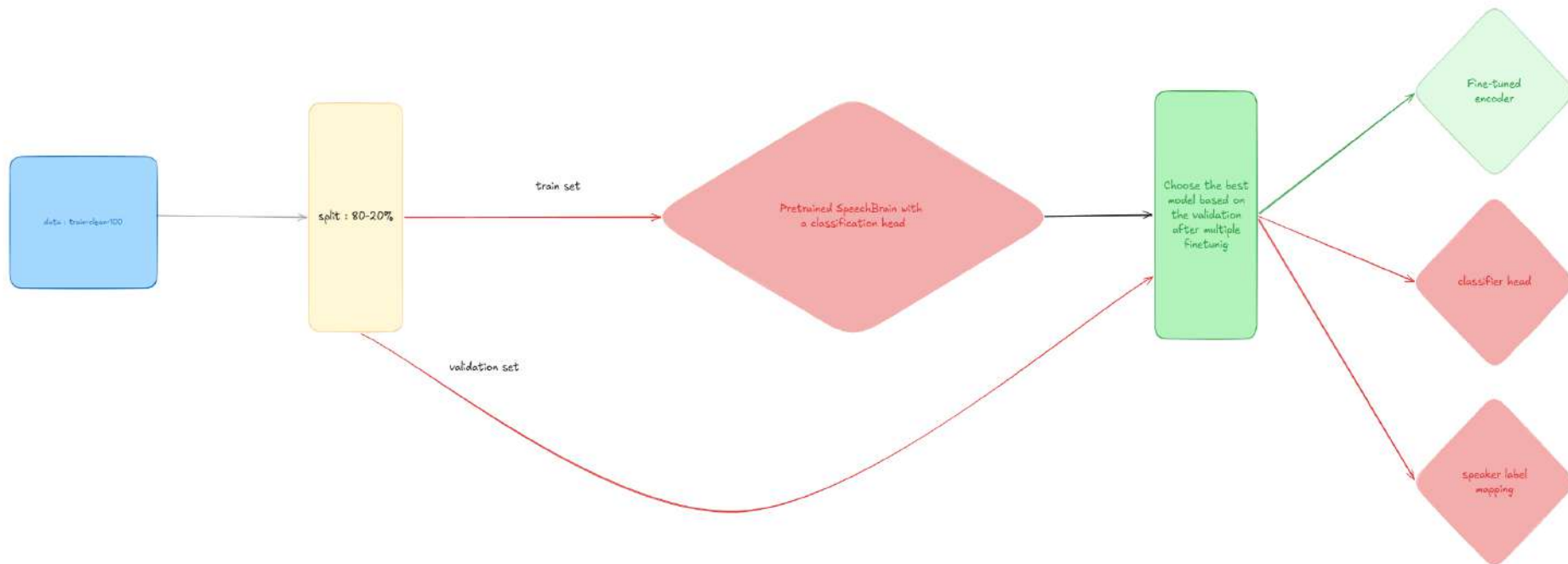
Great! I'll bring the presentation on our latest results.

2. Overall System Architecture



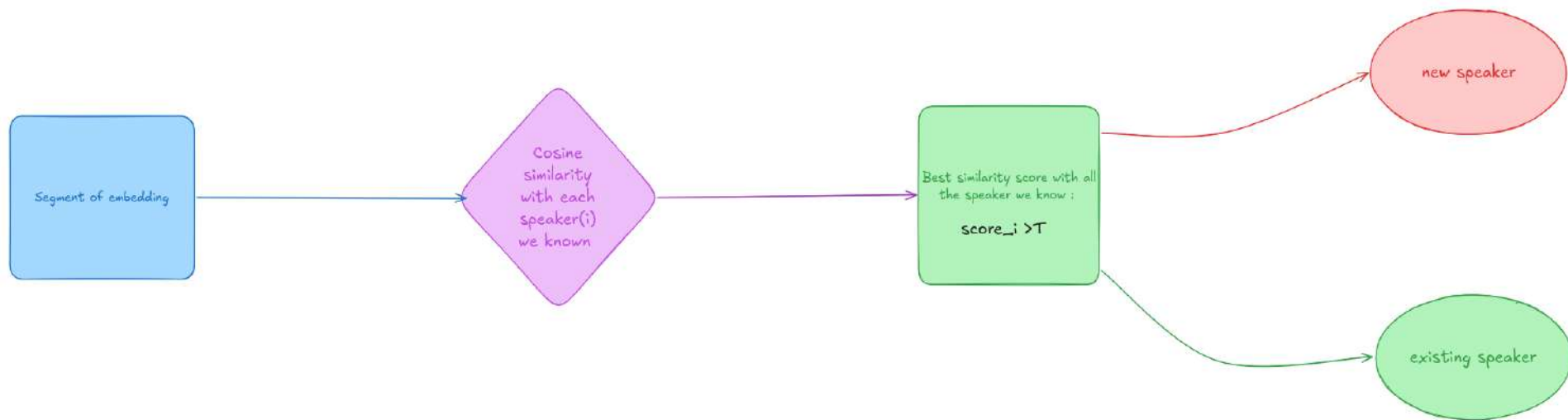
3. Model Setup and Preparation

Fine-tuned encoder



3. Model Setup and Preparation

Diarizer



4. API Design & Model Serving

Flask + Gunicorn

ENDPOINTS

POST /transcribe

Audio → JSON (segments, timestamps, speakers)

GET /health

Cloud Run readiness probe

GET /past_transcriptions

In-memory session history

PROCESSING PIPELINE

1

Audio Input

Multipart upload → temp file

2

FFmpeg Decode

PCM 16 kHz mono

3

Vosk ASR

Word-level timestamps

4

ECAPA-TDNN

192-dim speaker embedding

5

Diarizer

Cosine similarity → speaker ID

6

JSON Output

Segments from HearEdit timeline



Web UI

Drag-and-drop upload • Live results table • .txt report via Blob API (no extra server request)

5. Deployment in the Cloud

Docker · Google Cloud Run · GCS Model Storage

Container Setup

Base image: python:3.11-slim + FFmpeg (apt)

Build: uv instead of pip — faster install

Models NOT baked in: avoids 5 GB image

Downloaded from GCS at container startup

GCS Model Bucket

Bucket: hearedit-models

Vosk model: ~2.7 GB

ECAPA weights: ~80 MB (.pt)

Cloud Run Config

RAM: 16 Gi (Vosk 2.7 GB + PyTorch/SpeechBrain ~5 GB)

min-instances: 0 (scale to zero)

max-instances: 1 (no parallel model loads)

Public URL: <https://hearedit-api-726024632692.europe-west1.run.app/>

RAM Budget — 16 Gi

Vosk (2.7 GB)

PyTorch + SB (5 GB)

OS (2 GB)

Free (6.3 GB)

Demonstration

Thank you

Questions?

HearEdit · Speaker Diarization · INFO9023 Spring 2026 · Milestone 2

